The Journal of Open Source Software

adapol: Adaptive pole-fitting for Matsubara functions

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DOI: 10.xxxxxx/draft

Software

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Editor:

Submitted: 12 January 2025
Published: unpublished

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Summary

The Green's function approach to quantum many-body physics aims to replace high-dimensional wavefunctions with correlation functions which are more closely related to experimental observables of interest, such as spectral and response functions. Within this framework, real-time quantities, such as the Green's function, self-energy, and hybridization functions, are often represented in the discrete "Matsubara" domain on the imaginary frequency axis. A variety of physical observables can be directly recovered from the Matsubara Green's function, and many quantities of interest can be calculated more efficiently in this formalism.

A common computational task within this framework is decomposing a Matsubara function into a sum of simple poles:

$$G_{ab}(\mathrm{i}\omega_n) \approx \sum_{p=1}^{N_p} \frac{v_{ap} v_{bp}^*}{\mathrm{i}\omega_n - E_p}.$$

Here, $\mathbf{G}(i\omega_n)$ is a matrix-valued function of the Matsubara frequency point $\mathrm{i}\omega_n = (2n+1)\pi\mathrm{i}/\beta$ for fermionic functions, and $i\omega_n = 2n\pi\mathrm{i}/\beta$ for bosonic functions, with β representing the inverse temperature and $n \in \mathbb{Z}$. The pair $(E_p, \mathbf{v}_p)$ can be interpreted as a set of eigenpairs for an effective non-interacting Hamiltonian model. This "pole-fitting" problem is a crucial step in various numerical methods, such as hybridization fitting for quantum impurity solvers (Gazizova et al., 2024; Georges et al., 1996; Kaye et al., 2024; Mejuto-Zaera et al., 2020; Shinaoka & Nagai, 2021), and the analytic continuation of Matsubara Green's functions (Fei, Yeh, & Gull, 2021; Fei, Yeh, Zgid, et al., 2021; Huang et al., 2023; Ying, 2022; Zhang et al., 2024; Zhang & Gull, 2024). As a consequence of the rank-one positive-semidefiniteness constraint implied by the model, a best fit from Matsubara frequency data cannot be obtained component-wise, leading to a highly non-convex optimization landscape (see, e.g., Fig. 3 in (Huang et al., 2023)). Additionally, given that Matsubara data may be noisy in certain applications, instabilities can arise in methods which rely solely on standard rational approximation techniques (Fei, Yeh, & Gull, 2021; Schött et al., 2016).

Our Python package `adapol` ("add a pole") implements an adaptive pole-fitting procedure introduced in (Huang et al., 2023; Huang, Kaye, et al., 2024). The method first uses the AAA rational approximation algorithm (Nakatsukasa et al., 2018) to find an initial guess for the pole locations E_p. It then uses non-convex optimization and singular value decomposition to refine E_p and obtain v_p. Variants of this procedure have been shown to provide an accurate and compact fit for Matsubara data in a black-box and noise-robust manner, enabling new algorithms for dynamical mean-field theory (Mejuto-Zaera et al., 2020) and Feynman diagram evaluation (Huang, Kaye, et al., 2024). For example, (Huang, Kaye, et al., 2024) demonstrates that the procedure yields a more compact pole approximation than the generic discrete Lehmann representation (Kaye et al., 2022) for fixed objective functions.

Statement of Need

adapol is a simple and self-contained package which can be incorporated into codes requiring Matsubara pole-fitting. It includes a specialized API for common tasks such as hybridization fitting and analytic continuation, along with a user-friendly interface to the TRIQS package (Parcollet et al., 2015), enabling TRIQS users to utilize adapol with minimal modifications to their existing code.

adapol is distributed under the Apache License Version 2.0 and is available on GitHub (Huang, Yeh, et al., 2024a). The documentation (Huang, Yeh, et al., 2024b) provides background on the physics of Matsubara functions and the mathematics of pole-fitting, along with a detailed user guide, example applications, the TRIQS interface, and API reference documentation for all functions.

Acknowledgements

This work is partially supported by the Simons Targeted Grants in Mathematics and Physical Sciences on Moiré Materials Magic (Z.H., L.L.). The Flatiron Institute is a division of the Simons Foundation.

Fei, J., Yeh, C.-N., & Gull, E. (2021). Nevanlinna analytical continuation. *Phys. Rev. Lett.*, *126*(5), 056402. https://doi.org/10.1103/PhysRevLett.126.056402

Fei, J., Yeh, C.-N., Zgid, D., & Gull, E. (2021). Analytical continuation of matrix-valued functions: Carathéodory formalism. *Phys. Rev. B*, *104*, 165111. https://doi.org/10.1103/PhysRevB.104.165111

Gazizova, D., Zhang, L., Gull, E., & LeBlanc, J. P. F. (2024). *Feynman diagrammatics based on discrete pole representations: A path to renormalized perturbation theories*. https://doi.org/10.48550/arXiv.2407.01389

Georges, A., Kotliar, G., Krauth, W., & Rozenberg, M. J. (1996). Dynamical mean-field theory of strongly correlated fermion systems and the limit of infinite dimensions. *Rev. Mod. Phys.*, *68*(1), 13. https://doi.org/10.1103/RevModPhys.68.13

Huang, Z., Gull, E., & Lin, L. (2023). Robust analytic continuation of Green's functions via projection, pole estimation, and semidefinite relaxation. *Phys. Rev. B*, *107*(7), 075151. https://doi.org/10.1103/PhysRevB.107.075151

Huang, Z., Kaye, J., Strand, H. U. R., & Golež, D. (2024). Efficient evaluation of bold strong coupling diagrams in imaginary time using hybridization fitting and separation of variables. *In Preparation*.

Huang, Z., Yeh, C., Lin, L., Kaye, J., & Wentzell, N. (2024a). Adapol: Adaptive pole fitting for quantum many-body physics. *GitHub Repository*. https://github.com/flatironinstitute/adapol

Huang, Z., Yeh, C., Lin, L., Kaye, J., & Wentzell, N. (2024b). Adapol: Adaptive pole fitting for quantum many-body physics. *GitHub-Hosted Documentation*. https://flatironinstitute.github.io/adapol

Kaye, J., Chen, K., & Parcollet, O. (2022). Discrete lehmann representation of imaginary time Green's functions. *Phys. Rev. B*, *105*(23), 235115. https://doi.org/10.1103/PhysRevB.105.235115

Kaye, J., Huang, Z., Strand, H. U., & Golež, D. (2024). Decomposing Imaginary-Time Feynman Diagrams Using Separable Basis Functions: Anderson Impurity Model Strong-Coupling Expansion. *Phys. Rev. X*, *14*(3), 031034. https://doi.org/10.1103/PhysRevX.14.031034

Mejuto-Zaera, C., Zepeda-Núñez, L., Lindsey, M., Tubman, N., Whaley, B., & Lin, L. (2020).

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Efficient hybridization fitting for dynamical mean-field theory via semi-definite relaxation. *Phys. Rev. B*, *101*(3), 035143. https://doi.org/10.1103/PhysRevB.101.035143

Nakatsukasa, Y., Sète, O., & Trefethen, L. N. (2018). The AAA algorithm for rational approximation. *SIAM J. Sci. Comput.*, *40*(3), A1494–A1522. https://doi.org/10.1137/16M1106122

Parcollet, O., Ferrero, M., Ayral, T., Hafermann, H., Krivenko, I., Messio, L., & Seth, P. (2015). TRIQS: A toolbox for research on interacting quantum systems. *Comput. Phys. Commun.*, *196*, 398–415. https://doi.org/10.1016/j.cpc.2015.04.023

Schött, J., Locht, I. L., Lundin, E., Grånäs, O., Eriksson, O., & Di Marco, I. (2016). Analytic continuation by averaging padé approximants. *Phys. Rev. B*, *93*(7), 075104. https://doi.org/10.1103/PhysRevB.93.075104

Shinaoka, H., & Nagai, Y. (2021). Sparse modeling of large-scale quantum impurity models with low symmetries. *Phys. Rev. B*, *103*, 045120. https://doi.org/10.1103/PhysRevB.103.045120

Ying, L. (2022). Analytic continuation from limited noisy Matsubara data. *J. Comput. Phys.*, *469*, 111549. https://doi.org/10.1016/j.jcp.2022.111549

Zhang, L., & Gull, E. (2024). Minimal pole representation and controlled analytic continuation of Matsubara response functions. *Phys. Rev. B*, *110*, 035154. https://doi.org/10.1103/PhysRevB.110.035154

Zhang, L., Yu, Y., & Gull, E. (2024). Minimal pole representation and analytic continuation of matrix-valued correlation functions. *Phys. Rev. B*, *110*, 235131. https://doi.org/10.1103/PhysRevB.110.235131

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